Rishit Dwivedi

Virginia Kovach

ENGL 2500 – 57

09 March 2026

Not All Chips Are Equal: The Energy and Access Politics of AI Hardware

NVIDIA CEO Jensen Huang has described AI as a "five-layer cake: energy, chips, infrastructure, models, and applications," emphasizing that advanced AI systems depend on a physical and technical stack, not algorithms alone. The processors that make modern deep learning possible include graphics processing units (GPUs), tensor processing units (TPUs), and field-programmable gate arrays (FPGAs). These accelerators differ significantly in performance, energy use, and cost, and those differences shape who can build AI systems and at what scale. This paper examines two related questions: which hardware accelerators are most energy-efficient for deep learning workloads, and whether increasingly specialized AI hardware primarily advantages large technology companies over smaller research institutions.

Hardware Accelerators designed for Deep Learning are specialized processors built to perform the enormous amount of mathematical operations performed by deep neural networks. To measure energy efficiency on the different Hardware Accelerators, energy efficiency is typically stated in terms of operations per watt. For a better understanding of what that means when you see it or read it, let's review this analogy. A car's fuel economy is measured in miles per gallon; the energy efficiency of a processor or Hardware Accelerator is measured as tera operations per second per watt (TOPS/W). The term "tera" means 1 trillion; an "operation" is a fundamental mathematical calculation (for example: multiplying two numbers), and a "watt" is a standard measurement unit for energy (electricity). So TOPS/W tells us how many trillions of calculations a chip can perform for each watt of electricity it consumes. A higher TOPS/W means the chip does more useful work with less energy. According to Silvano et al., these accelerators "efficiently handle the heavy computational workload of DNN algorithms," and are deployed across "ultra-low-power… devices to high-performance servers… and large-scale data centers" (Silvano et al. 3)[1]. This wide range of deployment — from a tiny sensor on a factory floor to a warehouse-sized data center — illustrates why energy efficiency matters so much. The chips powering AI are not confined to a single setting; they appear everywhere modern computing operates, and the energy costs compound at every scale.

To understand the debate over energy efficiency and access in AI computing, it is first necessary to understand what hardware accelerators are and why they became essential to modern deep learning. A deep learning model is, at its core, a massive collection of mathematical equations — primarily matrix multiplications, which are operations that combine rows and columns of numbers in structured grids. Training a model means running these calculations billions of times, adjusting the numbers slightly after each pass so the model gradually learns to recognize patterns, whether in images, speech, or text. Traditional CPUs (Central Processing Units) — the general-purpose processors found in every personal computer — execute instructions one at a time, or in small batches, which makes them far too slow for the scale of computation deep learning demands. Hardware accelerators solve this problem by performing many calculations at the same time, a technique called "parallel processing." As Silvano et al. explain, "The technological development of the last ten years significantly increased the computing power of GPUs, which, due to their highly parallel nature, are incidentally very well suited to accelerate neural network training algorithms" (Silvano et al. 5)[1]. In other words, GPUs were originally designed for rendering video game graphics — which also involves doing many small math operations at once — and researchers discovered that this same architecture could be repurposed to speed up deep learning. This accidental fit between graphics hardware and AI calculations is what launched the modern deep learning revolution.

The first artificial intelligence (AI) accelerator was the graphics processing unit (GPU), but now there are alternative options, such as the tensor processing unit (TPU) from Google. The basic difference between the two types of chips is flexibility versus energy efficiency for AI tasks. A GPU is a general-purpose chip and can perform many different types of processing (e.g., video rendering, crypto currency mining, physics simulations, and neural network training). To support all of the functions that a GPU can handle, the general-purpose chip must contain hardware for functions that may not be needed for any given task. In contrast, a TPU is an application-specific integrated circuit (ASIC) designed specifically to perform one type of processing very well: matrix multiplication. The heart of a TPU is the systolic array, a grid of small processing units. The data processed by the systolic array flows in waves from one processor (node) to the next in rows and columns, so that each processor passes its results to the next processor without having to write the result back to memory first. This arrangement reduces the amount of power used in moving data, which is one of the largest costs of energy consumption in modern computing.The result is a chip that is narrowly specialized but remarkably efficient within its domain. Silvano et al. report that “Their last TPU v4 implementation outperforms the previous TPU v3 by 2.1x and improves performance/Watt by 2.7x” (Silvano et al. 7)[1]. This means that each new generation of TPU does not just get faster; it gets faster *while using less energy per calculation*, a combination that GPUs have historically struggled to achieve because their general-purpose design makes energy optimization harder.

The energy efficiency gap between GPUs and TPUs will depend on the specific workload being run, as empirical benchmark comparisons have shown significant differences between both types of processors in their efficiencies while performing different tasks. For example, a study by Wang et al. compared several different systems on multiple deep learning workloads (e.g., image recognition via convolutional neural networks - CNN's) and deep learning tasks such as language translation (using transformers) and speech recognition (using deep speech 2). Their findings show that "The benefit of performance by switching from V100 to TPU is 1.1×–1.7× on the non-recurrent translation model and 1.5×–2.7× on CNNs" (Wang et al. 3)[2]. In simpler terms, TPUs outperformed NVIDIA's top GPU at the time, the Tesla V100, by anywhere from 10% to 170% depending on the type of model being trained. The wide range is important: TPUs excel at tasks dominated by dense matrix operations, like image recognition, but their advantage narrows for tasks with irregular computation patterns, like certain types of language processing. Wang et al. also found that among GPUs, the NVIDIA V100 achieved up to 5.2 times lower energy consumption than competing GPUs from AMD, thanks to its specialized Tensor Cores — small matrix-multiplication units embedded inside the GPU that mimic, on a smaller scale, what the TPU does across its entire chip (Wang et al. 12)[2]. These benchmarks make clear that no single accelerator is universally superior. The "best" chip depends entirely on the specific task being performed.

FPGAs occupy a distinct position in the accelerator landscape — less powerful than GPUs or TPUs for large-scale training, but uniquely energy-efficient for inference tasks at the edge. "Inference" refers to the phase after a model has already been trained, when it is deployed to make predictions on new data — like when a voice assistant recognizes your speech or a security camera identifies a face.Edge computing enables organisations to make predictions at a location (e.g., a camera, sensor, or autonomous vehicle) rather than sending data to a remote data centre for processing. A unique feature of FPGAs is that they can be reconfigured to achieve optimal performance with a specific neural network. A fixed-function chip (i.e., a GPU or TPU) has a pre-determined design, whereas an FPGA can be reprogrammed at the hardware level to provide the ideal configuration for a specific task. By matching up every transistor on an FPGA chip to its intended purpose, the wasted power associated with unused circuits can be removed from the design. Urías Jiménez explains that "FPGAs use less power than other processors… their high efficiency helps mitigate cooling costs, supporting the development of 'greener' AI technologies" (Urías Jiménez 3). In a concrete example, a railway inspection system built on a Xilinx FPGA achieved energy efficiency 1.39 times better than a comparable GPU and 4.67 times better than a CPU, while processing 60 images per second with latencies below 17 milliseconds (Urías Jiménez 3). However, FPGAs come with significant trade-offs. Programming them requires specialized hardware-design expertise that most software engineers do not have, and their raw computational power is far lower than that of GPUs or TPUs, making them unsuitable for training large models. They are best understood as a complement to, rather than a replacement for, the dominant accelerators.

The energy efficiency debate cannot be separated from a deeper question of access: the most efficient accelerators — TPUs in particular — are largely unavailable to researchers outside of a handful of powerful technology companies. GPUs can be purchased on the open market by anyone with the budget, and NVIDIA's CUDA software ecosystem has made them relatively straightforward to program. TPUs, however, exist almost exclusively within Google's cloud infrastructure. Researchers can rent time on TPUs through Google Cloud, but they cannot buy, own, or modify the hardware themselves. This architectural lock-in gives Google a structural advantage: it controls not only the hardware but also the software stack (TensorFlow) that is most optimized to run on it. FPGAs, while commercially available, require hardware design expertise that most AI researchers lack, creating a different but equally significant barrier. The result is a hardware landscape where the most energy-efficient options are either proprietary (TPUs) or prohibitively specialized (FPGAs), leaving GPUs as the de facto standard for researchers who want flexible, accessible hardware — even though GPUs are often the least energy-efficient option for many tasks.

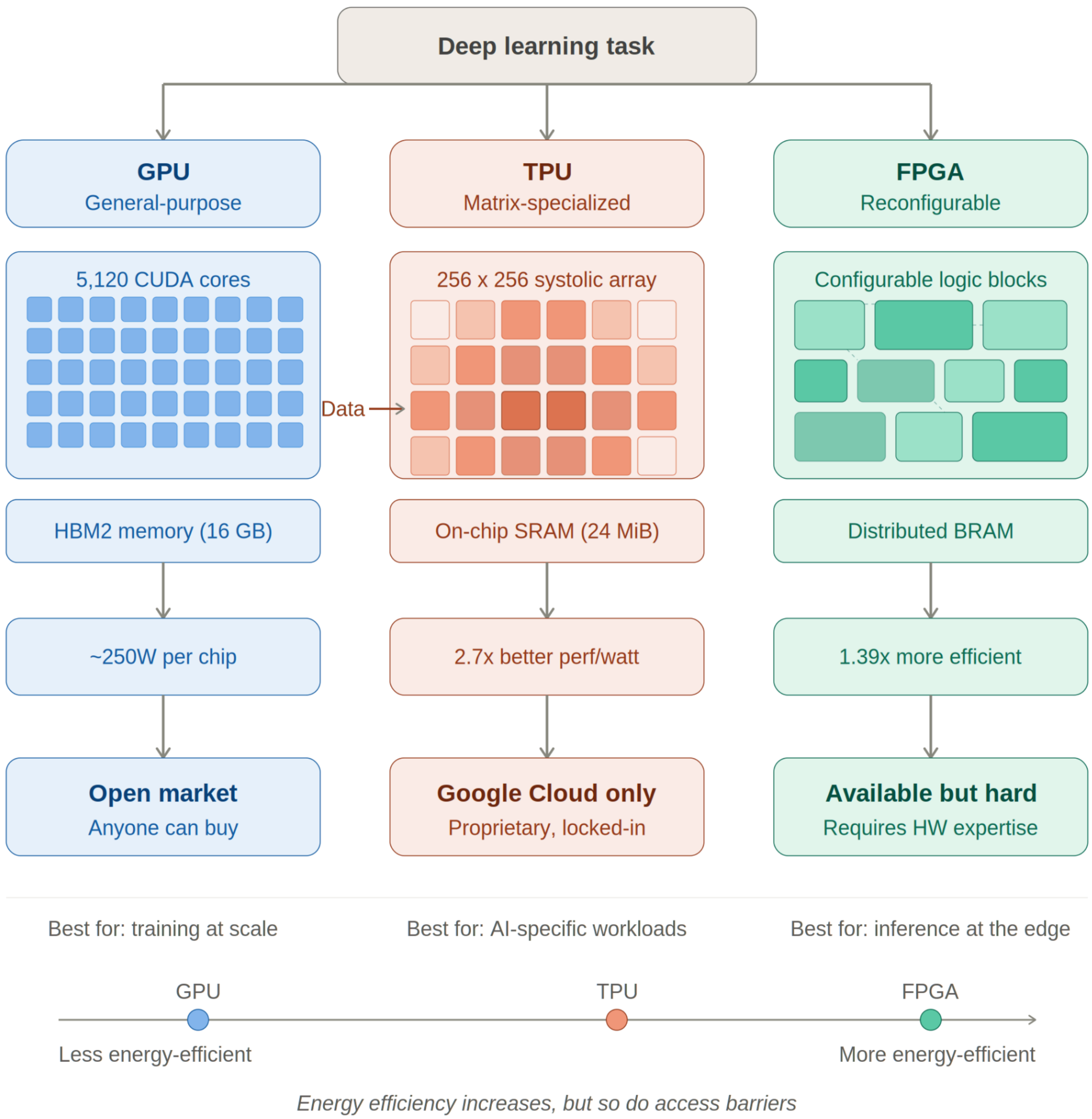

Fig. 1. Architectural comparison of GPU, TPU, and FPGA accelerators for deep learning.
Sources: Silvano et al. (2025); Wang et al. (2019); Urias Jimenez; Jouppi et al. (2017)

As illustrated in Figure 1, each accelerator type occupies a distinct position in the trade-off between energy efficiency and accessibility

The financial costs of cutting-edge AI hardware further entrench this access gap, creating a divide between well-funded tech giants and smaller research institutions. The Kak and West AI

Now Institute report makes this disparity vivid: “Some estimates suggest it will cost 3 million dollars a [month] and 20 million dollars in computing costs to train Pathways Language Model… Smaller businesses or start ups may consequently struggle to successfully enter this field, leaving the immense processing power of LLMs [in the hands of a few]” (Kak and West 34)[4]. These numbers are not abstractions; they represent real barriers that determine who gets to build AI and who does not. A university research lab operating on a typical federal grant of a few hundred thousand dollars per year simply cannot compete with a company that spends twenty million dollars training a single model. The concentration goes beyond training costs: the most powerful hardware — the latest NVIDIA H100 GPUs, Google’s TPU v4 pods — are allocated first to the companies that manufacture them or that have the purchasing power to secure early access. Larger companies have the ability to use the latest technology. To carry out their large experiments, smaller organizations must either use older, less capable technology or rent cloud computing power for a cost of $50-$100 an hour, which can be prohibitively expensive if done on a regular basis. This creates a self-perpetuating cycle because the companies that dominate the development of artificial intelligence (AI) are also the only ones with sufficient financial resources to purchase the necessary technology and remain on the cutting edge of innovation. In addition, smaller organizations in countries that have been historically marginalized—such as universities, non-profit organizations and start-up businesses in the Global South—are excluded from participating in the development of AI-based technologies that will have a profound impact on their societies. Ultimately, the concentration of AI hardware in a few large companies not only affects the ability of organizations to train AI models, but also determines the types of problems that AI is developed to address and the values that are reflected in the design of AI applications. Since Google, Microsoft, and Meta are the leaders in building large-scale AI systems, these systems focus on their business models: ads, recommending content to users, and offering services to consumers. However, many issues with AI such as climate change modeling and maintaining indigenous languages are often deferred to when there is access to computers and data by not-for-profit or public organizations for such long-term projects. The AI Now Report shows how this lack of access to AI computers extends into public policy too. For example, proposed ways to “democratize” AI through programs like the NAIRR have relied solely on licenses from the Big Tech Companies that have control over all aspects of the underlying computing infrastructure rather than working to create independent 3rd party resources to serve this purpose. Consequently, there are now many instances of government-funded programs at all levels of government creating the exact same kind of economic dependency on the existing status quo that is preventing democratization of AI through public investment into the future development of these technologies. The underlying computing layer for AI is not simply a technical platform; it is also a power source; thus, the design of chips used to create computers has political implications associated with energy and access; therefore, these implications will certainly affect all applications built upon them.

In this study, we evaluated a number of AI hardware accelerators, including GPUs, TPUs and FPGAs, based on two criteria: energy efficiency and accessibility to the hardware resources in an institution. For large training jobs, GPUs have superior performance across the board, however they are also extremely energy inefficient. TPUs are designed specifically for AI and demonstrate many of the advantages of GPUs, but utilise considerably less energy (better performance per watt) because they are limited to the Google data centers. FPGAs have the potential to be more energy efficient for edge inference tasks, while still having the ability to provide similar processing capabilities as a GPU (but not as much as TPUs) for larger model training.In conclusion, there is no one all-around excellent AI accelerator; rather the best accelerator will depend on task and use case. In regards to research question 2, the more powerful and energy-efficient hardware is concentrated among a handful of large technology companies; small research laboratories and start-ups have significant financial barriers (in the tens of millions of dollars) to build and train one model and/or to conduct their research. Taken together, the evidence shows that while no single accelerator is universally best, the most energy-efficient options are structurally concentrated in the hands of a few firms. This debate goes beyond computer engineering. When the most capable AI hardware is accessible only to Google, Microsoft, and a handful of other giants, the questions of who builds AI, whose problems it solves, and whose values it encodes are answered by default — not by choice. Hardware is not a neutral technical detail. It is the foundation upon which the entire AI ecosystem is built, and right now, that foundation is unequal. Understanding the energy and access politics of AI chips is the first step toward demanding something better.

Works Cited

Silvano, Cristina, et al. “A Survey on Deep Learning Hardware Accelerators for Heterogeneous HPC Platforms.” *ACM Computing Surveys*, vol. 57, no. 11, June 2025, article 286, pp. 1–39. https://doi.org/10.1145/3729215.

Wang, Yuxin, et al. “Benchmarking the Performance and Energy Efficiency of AI Accelerators for AI Training.” *arXiv*, 2019, arXiv:1909.06842.

Urías Jiménez, Arturo. "Beyond the GPU: The Strategic Role of FPGAs in the Next Wave of AI." *arXiv*, 4 Nov. 2025, arXiv:2511.11614.

Kak, Amba, and Sarah Myers West. “AI Now 2023 Landscape: Confronting Tech Power.” AI Now Institute, 11 Apr. 2023, https://ainowinstitute.org/2023-landscape.

Jouppi, Norman P., et al. “In-Datacenter Performance Analysis of a Tensor Processing Unit.” *ISCA ’17: Proceedings of the 44th Annual International Symposium on Computer Architecture*, ACM, June 2017, pp. 1–12. https://doi.org/10.1145/3079856.3080246.